Data Science Lab

Default of Credit Card Clients Prediction Model

Group 2

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GitHub Link for the Entire Project

SCAN ME

DIRECT LINK

Research Question and Hypotheses

To what extent can credit card default be predicted using repayment history , credit exposure, and borrower demographics?

1 Borrower characteristics associated with greater financial stability (eg. education and age) reduce the probability of default.

- age -> more experience and more savings -> less probability of default - marriage can help mitigate the default due to common savings
- education -> more financial awareness -> less probability of default

2 Poor repayment history (measured through past payment delays) increases the probability of credit default.

- diligent payments could signal a low probability of default

3 Higher credit exposure decreases the probability of default.

- higher limits are granted to more creditworthy borrowers

Dataset

Kaggle Dataset

Contains information on default payments, demographic factors, credit data, history of payment, and bill statements of credit card clients

Time & Place

Taiwan from April 2005 to September 2005

Variables

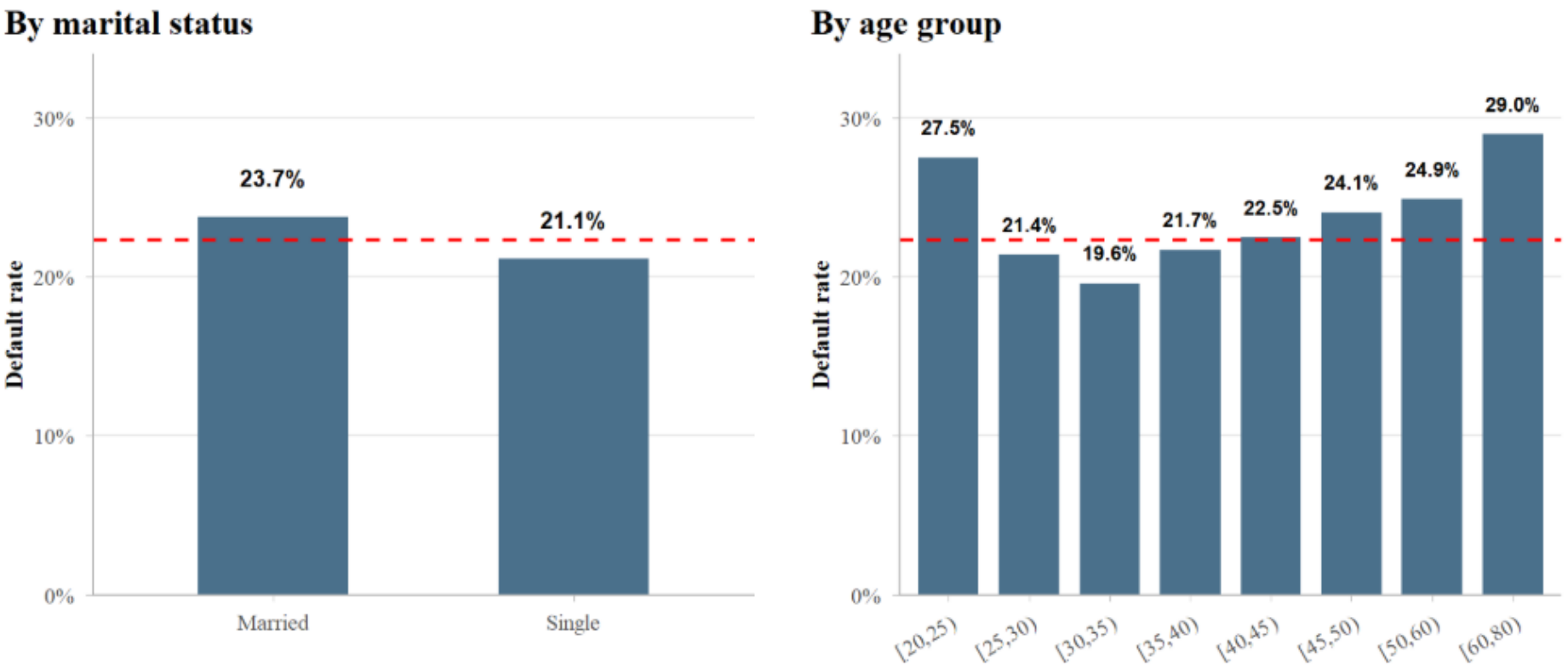
ID: ID of each client
LIMIT_BAL: Amount of given credit in NT dollars
SEX: Gender
EDUCATION
MARRIAGE: Marital status
AGE: Age in years
PAY_*X*: Repayment status in ***X***/2005
BILL_AMT*X*: Amount of bill statement in ***X***/2005
PAY_AMT*X*: Amount of previous payment in ***X***/2005
DEFAULT: Credit Default dummy

Visual Exploration

Dataset Visualization: Demographics and Default Rate

Figure 1: Default rates by demographic characteristics

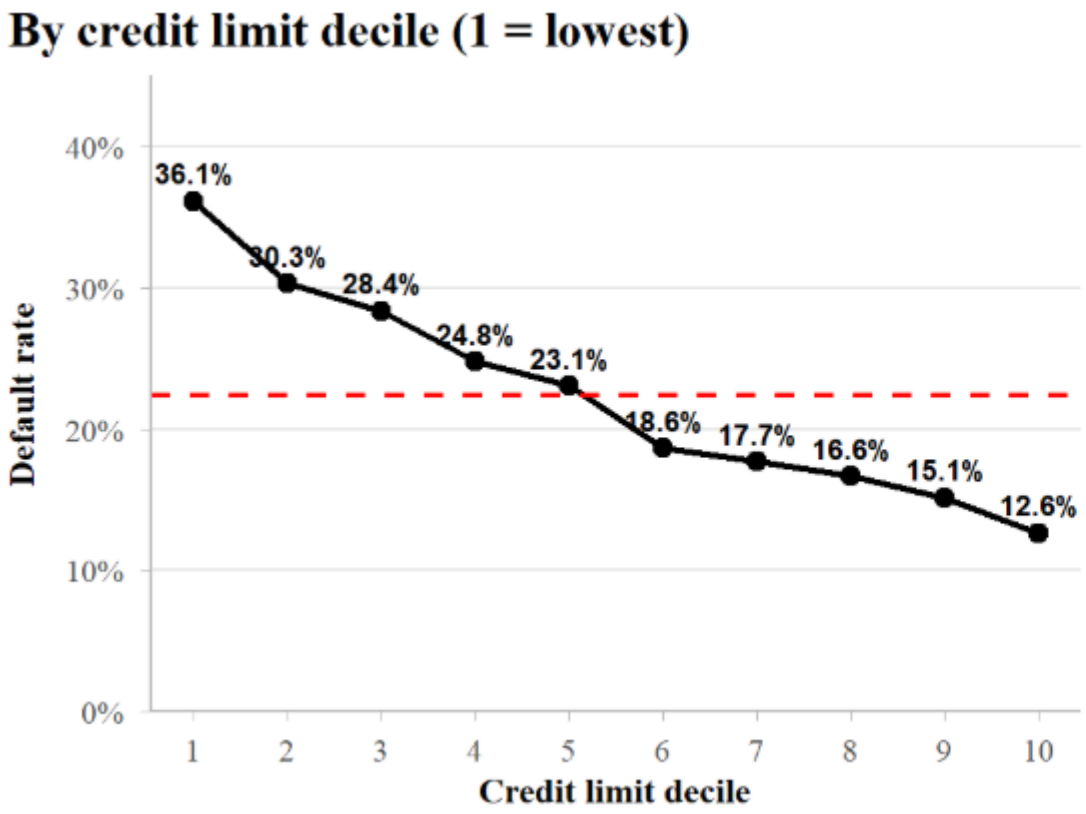
Dashed red line = overall default rate (22.1%)

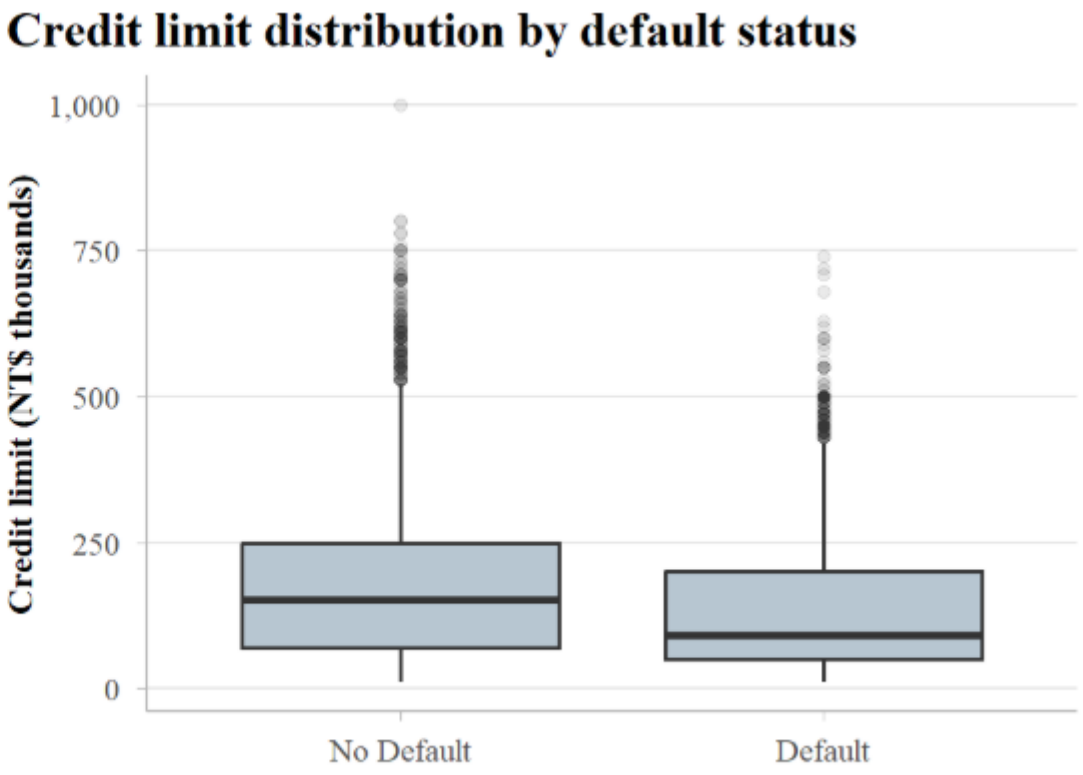


Dataset Visualization: Repayment History and Credit Exposure

Figure 2: Default rates by repayment behavior and credit exposure

Dashed red line = overall default rate (22.1%)





Relationships

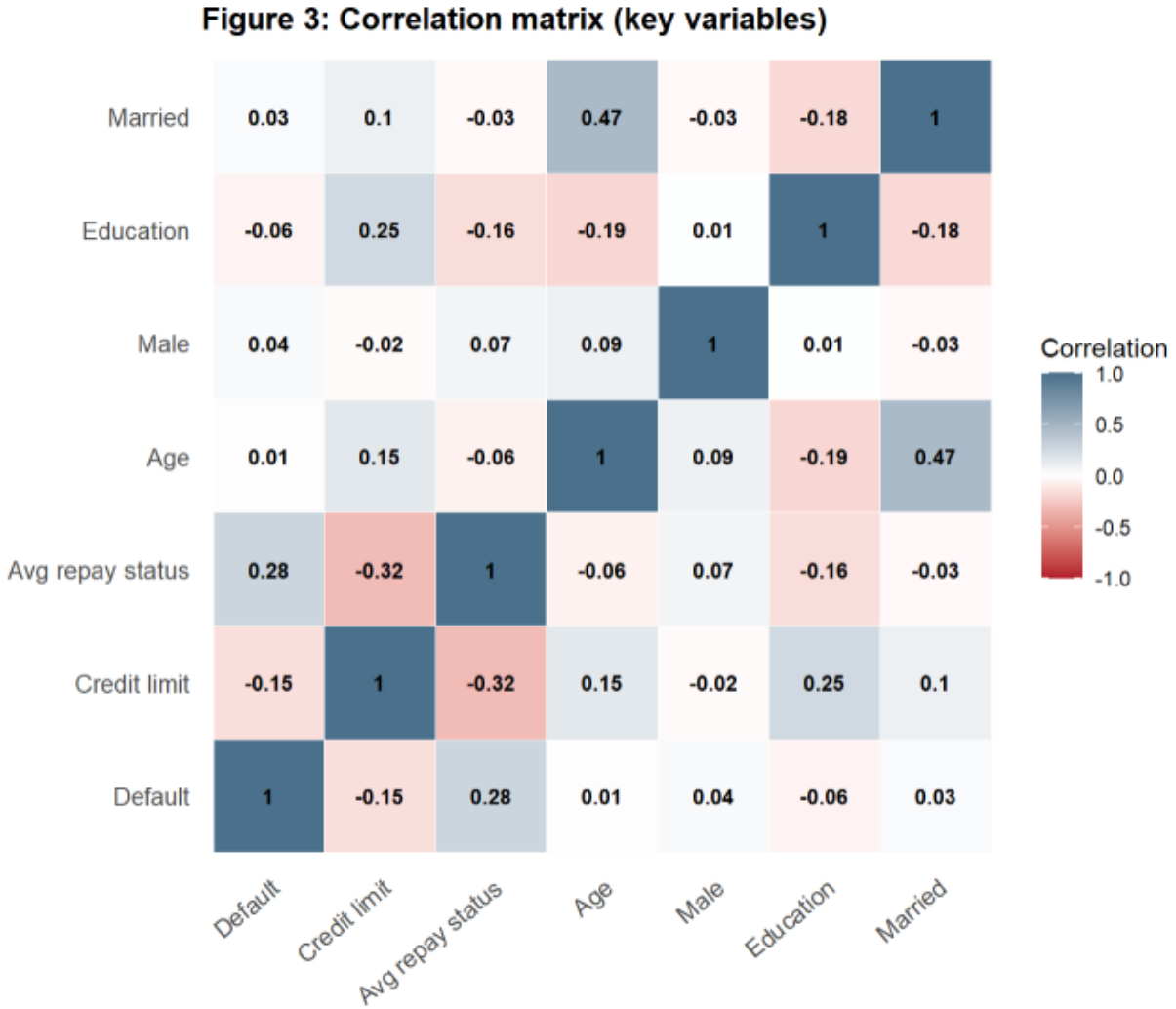


Figure 3: Pairwise correlation matrix of the outcome variable and key model predictors.

Empirical Study

Nested Logistic Regression Models

M1 Includes only pay_0 as a predictor, establishing a minimal baseline.

M2 Adds months_late, capturing the cumulative payment history.

M3 Adds limit_bal_th as a measure of credit exposure.

M4 Adds a full set of demographic controls:
age, age_sq, sex_male, education_level, and married.

Table 2: Logistic regression models — coefficient estimates

	M1	M2	M3	M4
(Intercept)	−1.394***	−1.728***	−1.506***	−1.461***
	(0.018)	(0.022)	(0.033)	(0.262)
pay_0	0.770***	0.348***	0.329***	0.329***
	(0.017)	(0.021)	(0.021)	(0.021)
months_late		0.393***	0.379***	0.376***
		(0.013)	(0.013)	(0.013)
limit_bal_th			−0.133***	−0.142***
			(0.015)	(0.017)
age				−0.013
				(0.014)
age_sq				0.000
				(0.000)
sex_male				0.052
				(0.037)
education_level				0.019
				(0.028)
married				0.157***
				(0.042)
Num.Obs.	21846	21846	21846	21846
AIC	20782.4	19871.6	19796.7	19774.8

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Standard errors in parentheses.

Regularized Logistic Regression

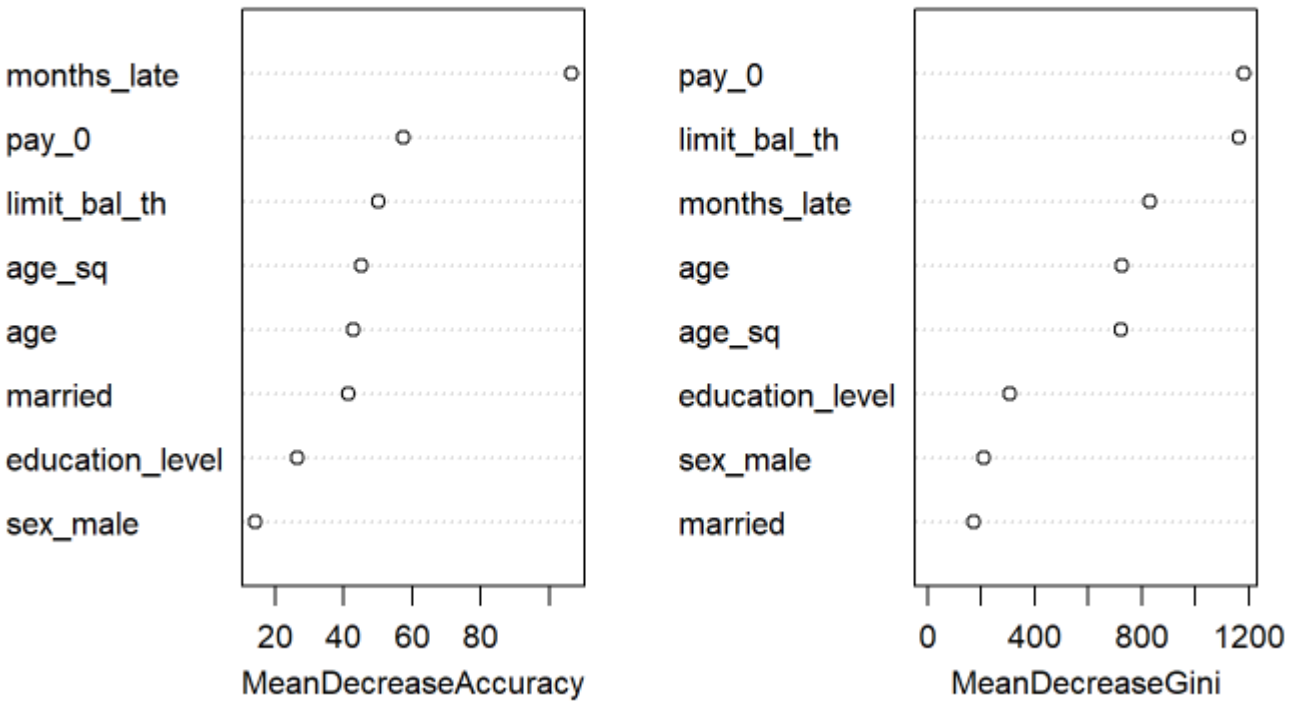
Table 3: Regularised regression coefficients at lambda.min

Variable	LASSO	Ridge
(Intercept)	-1.6337	-1.6338
pay_0	0.3238	0.3193
months_late	0.3751	0.3474
limit_bal_th	-0.1349	-0.1341
age	—	0.0007
age_sq	0	0.0000
sex_male	0.0348	0.0500
education_level	—	0.0033
married	0.1281	0.1271

LASSO lambda.min = 0.0012 | Ridge lambda.min = 0.0171. '—' indicates coefficient shrunk to exactly zero by LASSO.

Random Forest

Figure 4: Random Forest — variable importance



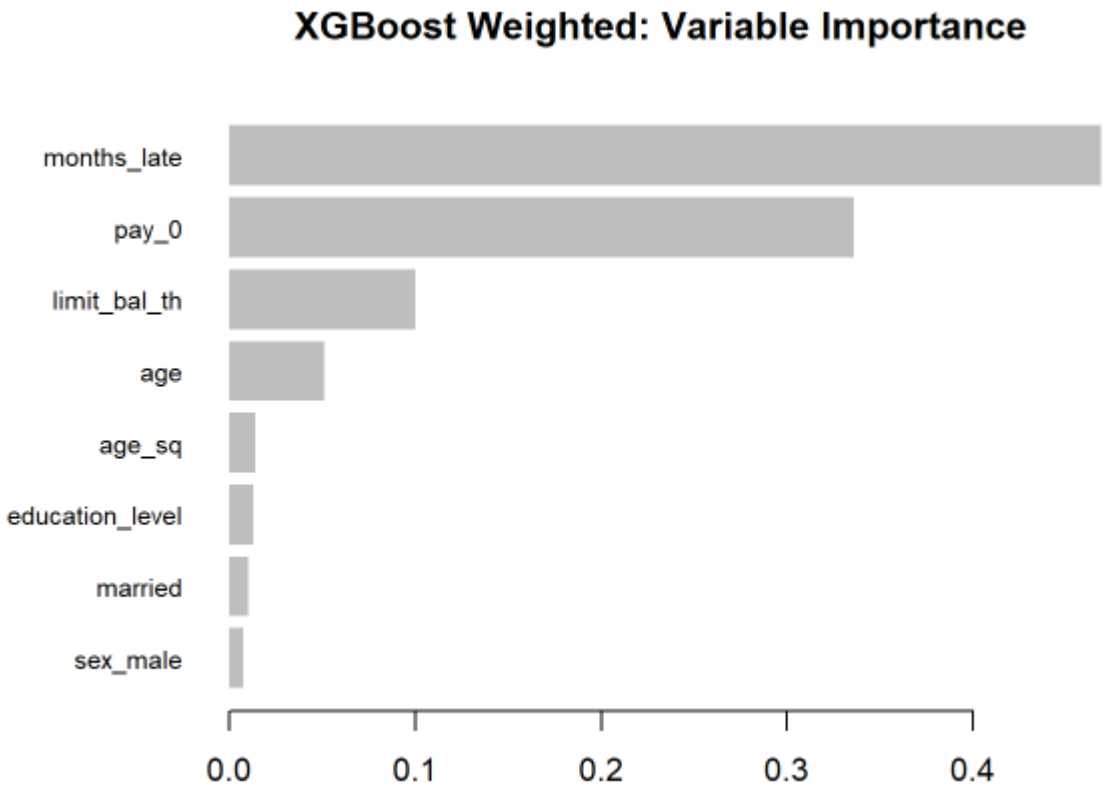
Gradient Boosting (XGBoost)

How it works

- Sequential trees (each corrects previous errors)
- Reduces bias (vs Random Forest → variance)

Handling class imbalance

- Class-weighted model (negative-to-positive class ratio ~3.5)
- Penalises missing defaulters → ↑ recall



Model Comparison

Model comparison on held-out test set

Model	AUC	Accuracy	Precision	Recall	F1
M1	0.6737	0.8157	0.6671	0.3145	0.4275
M2	0.7080	0.8044	0.6209	0.2724	0.3787
M3	0.7186	0.8044	0.6237	0.2674	0.3743
M4	0.7217	0.8036	0.6176	0.2687	0.3745
LASSO	0.7210	0.8036	0.6186	0.2668	0.3728
Ridge	0.7211	0.8043	0.6280	0.2586	0.3664
Random Forest	0.7253	0.7998	0.5711	0.3402	0.4264
XGBoost	0.7513	0.8127	0.6285	0.3515	0.4509
XGBoost Weighted	0.7518	0.7517	0.4481	0.5832	0.5068

ROC curves

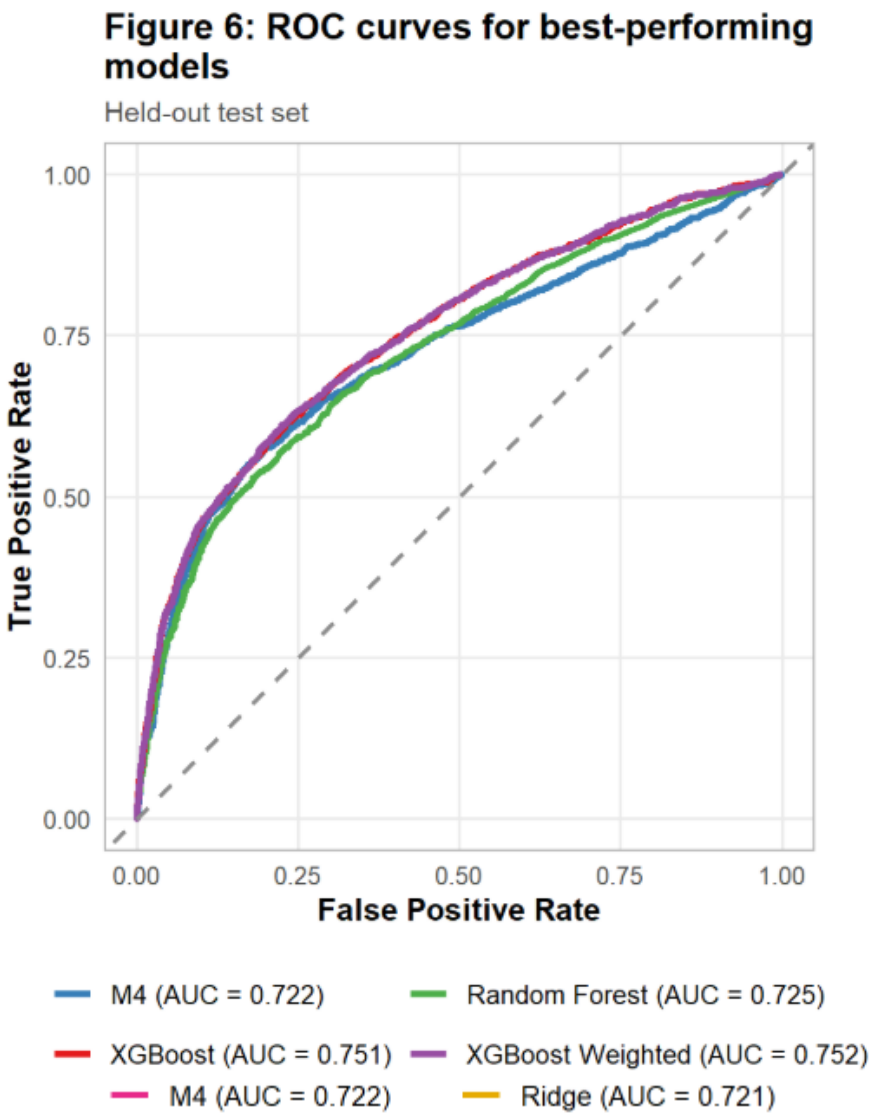


Figure 6: ROC curves for best-performing models
Held-out test set
True Positive Rate
False Positive Rate
0.00
0.25
0.50
0.75
1.00
M4 (AUC = 0.722)
Random Forest (AUC = 0.725)
XGBoost (AUC = 0.751)
XGBoost Weighted (AUC = 0.752)
M4 (AUC = 0.722)
Ridge (AUC = 0.721)

Results

Interpretation and Application

1. Repayments behaviour is the dominant signal for credit default prediction
2. Higher credit limits are robustly associated with lower default probability
3. Borrower demographics add little to prediction once behavioural and financial variables are controlled for, except for marital status. Married borrowers defaulting at higher rates is a robust but difficult-to-interpret finding, likely reflecting endogeneity and omitted variable bias.
4. Tree-based methods modestly outperform logistic regression.
5. The optimal threshold in practice depends on the bank's relative cost of false negatives versus false positives and should be calibrated accordingly rather than fixed arbitrarily, like in our case at 0.5.

Utrecht University

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